

Problem

Given a linear system with output $y(t)$ and impulse response $h(t)$, find the corresponding input $x(t)$ for the following cases:

(a) $y(t) = te^{-at}u(t)$, $h(t) = e^{-at}u(t)$

(b) $y(t) = u(t+1) - u(t-1)$, $h(t) = \delta(t)$

(c) $y(t) = e^{-at}u(t)$, $h(t) = \text{sgn}(t)$

Step-by-step solution

Step 1 of 3

(A) $y(t) = te^{-at}u(t)$

$h(t) = e^{-at}u(t)$

$$Y(\omega) = \frac{1}{(j\omega + a)^2}$$

$$H(\omega) = \frac{1}{(j\omega + a)}$$

$$\Rightarrow X(\omega) = \frac{\frac{1}{(j\omega + a)^2}}{\frac{1}{(j\omega + a)}} = \frac{1}{j\omega + a}$$

$$\left[H(\omega) = \frac{Y(\omega)}{X(\omega)} \right]$$

$$\Rightarrow \boxed{x(t) = e^{-at}u(t)}$$

Step 2 of 3

(B) $y(t) = u(t+1) - u(t-1)$

$h(t) = \delta(t)$

$$Y(\omega) = \frac{2}{j\omega}e^{j\omega} - \frac{2}{j\omega}e^{-j\omega}$$

$h(\omega) = 1$

$$\Rightarrow X(\omega) = \frac{\frac{2}{j\omega}e^{j\omega} - \frac{2}{j\omega}e^{-j\omega}}{1} = \frac{2}{j\omega}e^{j\omega} - \frac{2}{j\omega}e^{-j\omega}$$

$$\Rightarrow \boxed{x(t) = u(t+1) - u(t-1)}$$

Step 3 of 3

(C) $y(t) = e^{-at}u(t)$

$h(t) = \text{sgn}(t)$

$$Y(\omega) = \frac{1}{j\omega + a}$$

$$H(\omega) = \frac{2}{j\omega}$$

$$\Rightarrow X(\omega) = \frac{\left(\frac{1}{j\omega + a} \right)}{\left(\frac{2}{j\omega} \right)} = \frac{2}{(j\omega + a)} \times \frac{j\omega}{2} = \frac{1}{2} \frac{j\omega}{j\omega + a}$$

$$= \frac{1}{2} \left[1 - \frac{a}{j\omega + a} \right]$$

$$\Rightarrow \boxed{x(t) = \frac{1}{2} [\delta(t) - ae^{-at}u(t)]}$$